

Standard TEM fixation protocol (August, 2017)

Fixes

sequential:

2.5% GA (30min), rinse in medium (2x), 1%Os (30min)

simultaneous: 2% Os → 5% GA → cells (30min) (on ice!)

near simultaneous:

5% GA → cells ← 2% Os (30min) (on ice!)

pellet gently (long time at low rpm, or alternative)
(prep agarose during fixes)

disturb cell pellet before adding fixes!

2min @ 6000rpm (medium/buffer)

2min @ 6000rpm (water)

3min @ 7000rpm (water)

Agarose enrobing + early dehydration

aliquot dense disrupted pellet in small volume onto
agarose, roll + dry with hair tool. Cover with agarose,
cut block, re-enrobe. 1-2mm cube

Move agarose block to EtOH:

50% EtOH (10min)

70% EtOH (10min or o/n stopping point)

Dehydration + infiltration

(prep resin in the meantime)

70% EtOH 10min

90% EtOH 10min

95% EtOH 10min

100% EtOH 10min

100% EtOH 10min

100% EtOH 10min

100% Acetone 10min

100% Acetone 10min

1/3 R/Ac 1h

2/3 R/Ac 1h

R1 o/n

R2 morning

R3 o/n

make labels, pour into moulds, cure at 70°C

Resin:	(g)	1/2 (g)	
812	18.3	9.15	
DDSA	10.05	5.02	
NMA	10.9	5.45	
DMP30	0.6	0.3	(use transfer pipette)
(minimise exposure to humidity!)			

materials (fix + dehydration)

- ☐ transfer pipettes (5-10)
- ☐ centrifuge
- ☐ spare eppies
- ☐ parafilm – 3-5 strips, and a square
- ☐ gloves
- ☐ bench cover
- ☐ ice bucket(s), ice
- ☐ pre-loaded eppies with medium or buffer for Os, GA
- ☐ water, 2 x 15mL tubes
- ☐ filtered medium or buffer, 2 x 15mL tubes
- ☐ box of p1000 tips
- ☐ p1000 pipette
- ☐ timer
- ☐ sharpie
- ☐ kimwipes
- ☐ cells
- ☐ 50% EtOH
- ☐ 70% EtOH

materials (dehydration)

- ☐ several mL agarose (pre-melted!)
- ☐ slides (coated with agarose)
- ☐ coverslips (for cutting)
- ☐ hair tools, transfer pipettes
- ☐ heating block + thermometer (to 80-90°C)

materials (dehydration)

- ☐ 50% EtOH
- ☐ 70% EtOH
- ☐ 90% EtOH
- ☐ 95% EtOH
- ☐ 100% (anhydrous!) EtOH
- ☐ 100% (anhydrous!) Acetone
- ☐ screwcap tubes
- ☐ rotator
- ☐ timer
- ☐ transfer pipettes

materials (resin)

- ☐ scale (cover with bench paper [])
- ☐ stir plate
- ☐ resin stir bar
- ☐ 2 large pieces of parafilm to cover beaker
- ☐ 2 small pieces of parafilm for syringe
- ☐ 4 pieces of parafilm for bottles
- ☐ extra parafilm
- ☐ 100mL plastic beaker
- ☐ 2x 15mL tubes of 100% acetone (for cleaning)
- ☐ 2x 60mL syringes
- ☐ box of kimwipes
- ☐ gloves
- ☐ rotator
- ☐ resin waste bag (labelled)
- ☐ wad of transfer pipettes
- ☐ curing oven
- ☐ molds